

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

Experiment 3

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Subject Name: Full Stack

Subject Code: 23CSH-382

1. Aim: To optimize the performance of the EcoTrack React application using memoization techniques and code splitting, and to enhance the user interface using enterprise-grade Material UI components.

2. Objective:

After completing this experiment, the student will be able to:

1. Understand the causes of unnecessary re-renders in React applications
2. Optimize React components using React.memo to prevent avoidable re-renders
3. Apply useMemo to efficiently compute derived data and avoid redundant calculations
4. Use useCallback to memoize event handler functions and improve component performance
5. Implement lazy loading of components and routes using React.lazy and Suspense
6. Reduce initial bundle size and improve application load performance through code splitting
7. Enhance the visual appearance and usability of the EcoTrack application using Material UI components
8. Design a clean, consistent, and responsive user interface using Material UI layouts and typography

3. Implementation/Code:

Header.js:

```
import {Link} from "react-router-dom";
```

```
const Header = ({ title }) => {  
  return (  

```

```

<header style= {{
  padding:"1rem",
  backgroundColor:"#27ae60",
  color:"white",
  textAlign:"center",
}}>
  <h2>{title}</h2>
  <nav>
    <Link to="/" style={{marginRight:"1rem",color:"#79e44f"}}>
      Dashboard
    </Link>
    <Link to="/logs"
style={{marginRight:"1rem",color:"#79e44f"}}>
      Logs
    </Link>
    <Link to="/login"
style={{marginRight:"1rem",color:"#79e44f"}}>
      Login
    </Link>

    <Link to="/logout"
style={{marginRight:"1rem",color:"#79e44f"}}>
      Logout
    </Link>

  </nav>
</header>
);
};

export default Header;

```

AuthContext.js:

```

import { createContext,useContext,useState } from "react";

const AuthContext = createContext(null);

export const AuthProvider =( {children} )=>{

```

```

const [ isAuthenticated, setIsAuthenticated ] = useState(false);

return(
  <AuthContext.Provider value = {{ isAuthenticated,
    setIsAuthenticated }}>
    {children}
  </AuthContext.Provider>
)
}

export const useAuth = () => useContext(AuthContext);

```

logs.js:

```

export const logs = [
  { id: 1, activity: "Car Travel", carbon: 4 },
  { id: 2, activity: "Electricity Usage", carbon: 6 },
  { id: 3, activity: "Cycling", carbon: 0 },
  { id: 4, activity: "Bus Travel", carbon: 3 },
  { id: 5, activity: "Solar Energy Usage", carbon: 1 },
  { id: 6, activity: "Flight Travel", carbon: 8 },
];

export default logs;

```

Dashboard.js:

```

import logs from '../data/logs'
import Header from '../components/header'
const totalCarbon = logs.reduce ((sum, log) => sum+log.carbon,0)

const Dashboard = () => {
  return (

    <div >
      <h2>{Header}</h2>
      <p style={{padding
        : "1rem",backgroundColor:""#3175a2",color:""#d6eaf8",textAlign:"center"
      }}>Total Carbon Footprint: {totalCarbon}</p>

      <ul style={{padding

```

```

      : "1rem", backgroundColor: "white", color: "#111", textAlign: "center" }} >
      {logs.map((log) => (
        <li key={log.id}>
          {log.activity}: {log.carbon} kgs
        </li>
      ))}
    </ul>
  </div>

)
}
export default Dashboard;

```

DashboardAnalytics.js:

```

const DashboardAnalytics = () => {
  return (
    <p>This is Dashboard Analytics</p>
  )
};
export default DashboardAnalytics;

```

DashboardSummary.js:

```

const DashboardSummary = () => {
  return (<p>This is Dashboard Summary</p>)
}

export default DashboardSummary;

```

DashboardSettings.js:

```

const DashboardSettings = () => {
  return (<p>These are the settings</p>)
}

export default DashboardSettings;

```

DashboardLayout.js:

```

import { Link, Outlet } from "react-router-dom";

```

```

const DashboardLayout = ()=>{

  return (
    <div style={{padding:"1rem"}}>
      <h3>Dashboard</h3>

      <nav>
        <Link to="/summary">Summary</Link> |{" "}
        <Link to="/analytics">Analytics</Link>
      </nav>
      <hr/>
      <Outlet/>

    </div>

  )
};

export default DashboardLayout;

```

Login.js:

```

import { useAuth } from "../context/AuthContext";
import { useNavigate } from "react-router-dom";

const Login =()=>{
  const {setIsAuthenticated}=useAuth();
  const navigate = useNavigate();

  const handleLogin=() => {
    setIsAuthenticated(true);
    navigate("/");
  };

  return (
    <div style={{padding:"1rem"}}>
      <h3>Login Page</h3>
      <button onClick={handleLogin}>Login</button>
    </div>
  )
}

```

export default Login;

Logout.js:

```
import { useAuth } from "../context/AuthContext";
import { useNavigate } from "react-router-dom";

const Logout = () => {
  const { setIsAuthenticated } = useAuth();
  const navigate = useNavigate();

  const handleLogout = () => {
    setIsAuthenticated(false);
    navigate("/");
  };

  return (
    <div style={{ padding: "1rem" }}>
      <h3>Logout Page</h3>
      <button onClick={handleLogout}>Logout</button>
    </div>
  )
}

export default Logout;
```

ViewLogs.js:

```
import logs from "../data/logs";

const Viewlogs = () => {
  return (
    <div>
      <h2>Experiment Logs</h2>
      <ul>
        {logs.map((log, index) => (
          <li key={log.id}>
            {log.activity}: {log.carbon} kg CO2
          </li>
        ))}
      </ul>
    </div>
  )
}
```

```
}  
export default Viewlogs;
```

PrototectedRoute.js:

```
import {Navigate} from "react-router-dom";  
import { useAuth } from "../context/AuthContext";
```

```
const ProtectedRoute= ({children})=>{  
  const {isAuthenticated} = useAuth();  
  
  if(!isAuthenticated){  
    return <Navigate to="/login" replace/>  
  }  
  
  return children;  
}
```

```
export default ProtectedRoute;
```

LogSlice.js:

```
import { createSlice,createAsyncThunk } from "@reduxjs/toolkit";
```

```
export const fetchLogs = createAsyncThunk(  
  "logs/fetchLogs",  
  async()=> {  
    await new Promise ((resolve)=>setTimeout(resolve,1000));  
    return [  
      { id: 1, activity: "Car Travel", carbon: 4 },  
      { id: 2, activity: "Electricity Usage", carbon: 6 },  
      { id: 3, activity: "Cycling", carbon: 0 },  
      { id: 4, activity: "Bus Travel", carbon: 3 },  
      { id: 5, activity: "Solar Energy Usage", carbon: 1 },  
      { id: 6, activity: "Flight Travel", carbon: 8 },  
    ];  
  }  
)
```

```
const logsSlice = createSlice({  
  name: "logs",  
  initialState: {  
    data:[],
```

```

        status:"idle",
        error:null,
    },
    reducers: {},
    extraReducers :(builder)=>{
        builder.addCase(fetchLogs.pending,(state)=>{
            state.status="loading";
        })
        .addCase(fetchLogs.fulfilled,(state,action)=>{
            state.status ="success";
            state.data = action.payload;
        })
        .addCase(fetchLogs.rejected,(state,action)=>{
            state.status = "failed";
            state.error = action.error.message;
        })
    }
})

```

```
export default logsSlice.reducer;
```

Store.js:

```

import {configureStore} from "@reduxjs/toolkit"
import logsReducer from "./logsSlice";

```

```

const store = configureStore({
  reducer :{
    logs: logsReducer,
  },
});

```

```
export default store;
```

App.js:

```

import Header from "./components/header";
import { Suspense, lazy } from "react";

```



```

import { BrowserRouter,Routes,Route } from "react-router-dom";

const Login = lazy(()=>import("./pages/login"))
const DashboardLayout = lazy(()=> import("./pages/DashboardLayout"))
const Viewlogs = lazy(()=>import("./pages/Viewlogs"))
const DashboardAnalytics =
  lazy(()=>import("./pages/DashboardAnalytics"))
const DashboardSummary =
  lazy(()=>import("./pages/DashboardSummary"))
const ProtectedRoute = lazy(()=>import("./routes/ProtectedRoutes"))
const Logout = lazy(()=>import("./pages/logout"))
const DashboardSettings =
  lazy(()=>import("./pages/DashboardSettings"))

function App () {
  return (
    <BrowserRouter>
      <Header title="Eco Track : Experiment 2 " />
      <Suspense fallback={<h2>Loading...</h2>}>
        <Routes>
          <Route path= "login" element={<Login/>} />
          <Route path= "/" element=
            {
              <ProtectedRoute>
                <DashboardLayout />
              </ProtectedRoute>
            }>

          <Route index />
          <Route path="settings" element={<DashboardSettings />} />
          <Route path="Summary" element={<DashboardSummary/>} />
          <Route path="Analytics" element={<DashboardAnalytics/>} />
          <Route path="logs" element={<Viewlogs/>} />

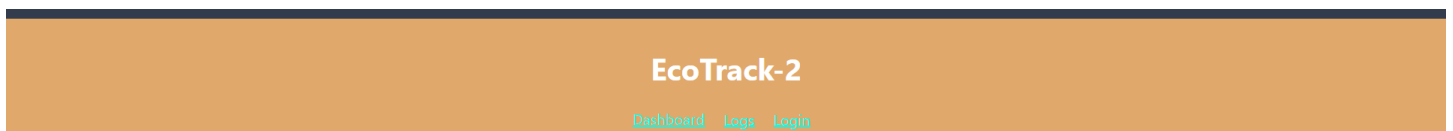
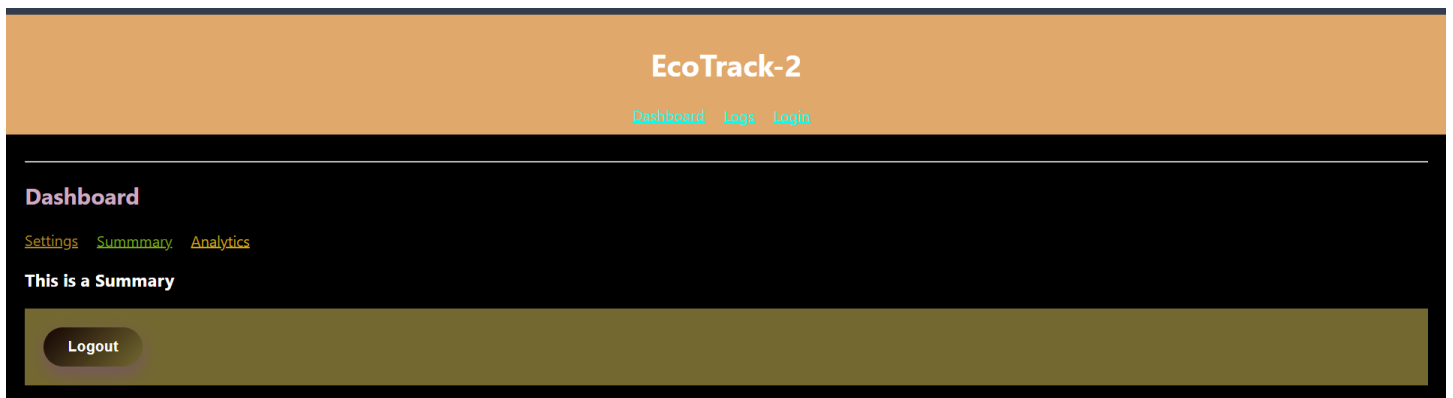
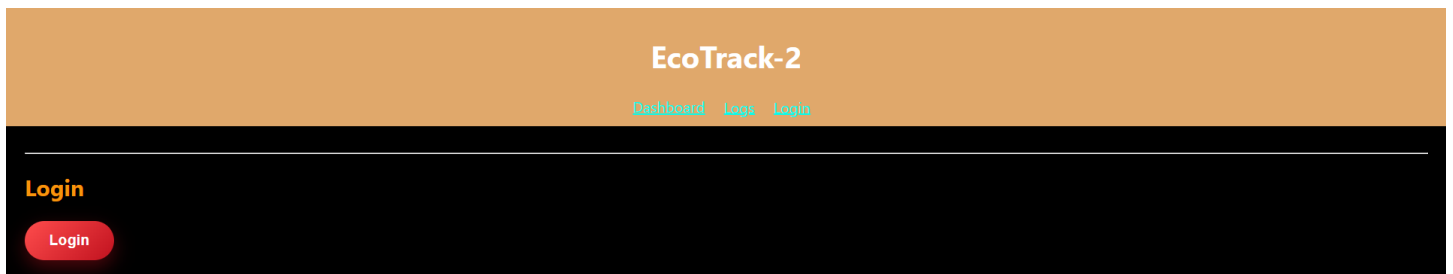
        </Route>
      </Routes>
    </Routes>
    <Route path="logout" element={<Logout/>} />
  </Routes>

```

```
</Suspense>
```

```
</BrowserRouter>  
)  
}  
export default App;
```

4. Output:



Daily Logs (Redux)

- car travel | 4 kg CO₂
- Electricity Usage | 2 kg CO₂
- Cycling | 0 kg CO₂

5. Learning Outcome

- 1. Learnt about react files.**
- 2. Learnt Design and apply redux and its uses.**
- 3. Learnt Implement nested routing to build dashboard-style layouts.**